

Telecommunication Assignment -3



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Task - 1

What is a periodogram? What is the advantage in comparison with the standard method? What's the benefit of Welch and Lomb-Scargle periodograms?

In signal processing, a periodogram is an estimate of the spectral density of a signal. Today, the periodogram is a component of more sophisticated methods. It is the most common tool for examining the amplitude vs frequency characteristics of FIR filters and window functions. This can be a helpful tool for identifying the dominant cyclical behaviour in a series.

$$PSD(f) = \frac{|FFT(w_t * X_t, f)|^2}{L} * 1/\Delta f$$

X_t is a time series

L is the number of samples in the time series

w_t is the applied window function

Δf is the frequency interval

We will represent the time series as

$$x_t = \sum_{j=1}^{n/2} \left[\beta_1 \left(\frac{j}{n} \right) \cos(2\pi w_j t) + \beta_2 \left(\frac{j}{n} \right) \sin(2\pi w_j t) \right]$$

This is a sum of sine and cosine functions at the harmonic frequencies. Think of the $\beta_1(j/n)$ and $\beta_2(j/n)$ as regression parameters. Then there are a total of n parameters because we let j move from 1 to $n/2$. This means that we have n data points and n parameters, so the fit of this regression model will be exact.

The advantage in comparison with the standard method:

- A periodogram calculates the significance of different frequencies in time-series data to identify any intrinsic periodic signals.
- A periodogram is similar to the Fourier Transform but is optimized for unevenly time-sampled data, and for different shapes in periodic signals.
- It is simple to compute while compared with other methods.

Benefits of Welch and Lomb-Scargle periodograms:

Welch's method is also called the periodogram method for estimating power spectra is carried out by dividing the time signal into successive blocks, forming the periodogram for each block, and averaging. In python by using the `scipy. signal. welch` estimates the power spectral density by dividing the data into segments and averaging the periodograms computed on each segment.

Benefits of the welch method is

- Its mainly improves the standard periodogram spectrum estimation by averaging over several FFT packets.
- It also reduces the noise in the estimated power spectra, in exchange for reducing the frequency resolution.

The Lomb-Scargle periodogram is a commonly used statistical tool designed to detect periodic signals in unevenly spaced observations. The Lomb–Scargle periodogram is a method that allows efficient computation of a Fourier-like power spectrum estimator from such unevenly sampled data, resulting in an intuitive means of determining the period of oscillation.

Benefits of Lomb–Scargle periodogram method is

- It handles the uneven data
- It also provides the high frequency resolution
- It also handles with the missing data.

Task - 2

Implement and demonstrate periodogram in Python for two different examples (time series, e.g. multiple sinusoidals) using Welch and Lomb-Scargle.

2.1 Welch method.

Script :

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import signal

t = np.linspace(0, 10, 1000)
x = 1*np.sin(2*np.pi*2*t) + 0.3*np.sin(2*np.pi*6*t)
y = np.linspace(0.1, 10, 1000)

f, Pxx = signal.welch(x, fs=1/(t[1]-t[0]), nperseg=256)

fig, (ax_t, ax_w) = plt.subplots(2, 1, constrained_layout=True)
ax_t.plot(t, x)
ax_t.set_xlabel('Time [s]')

ax_w.plot(f, Pxx)
ax_w.set_xlabel('Angular frequency [rad/s]')
ax_w.set_ylabel('Normalized amplitude')
plt.show()
```

Output :

The noise power on the signal :

0.0009991424981818208

The peak height in the power spectrum :

2.0148191844362335

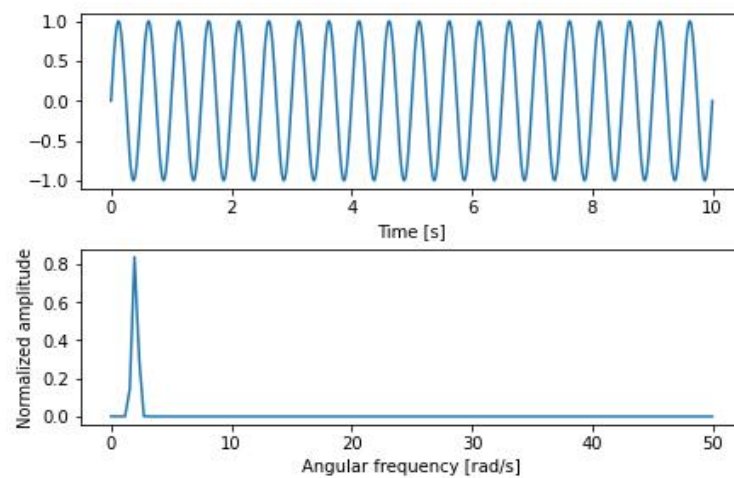


Fig 2.1.1 : Periodogram plot using welch for a single signal.

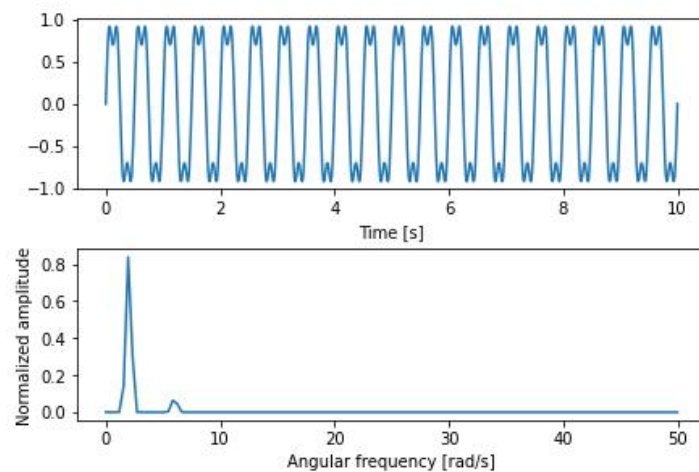


Fig 2.1.2 : Periodogram plot using welch for the multiple signal.

2.2 Lomb-Scargle.

Script :

```
import numpy as np
import matplotlib.pyplot as plt
import scipy.signal as signal
rng = np.random.default_rng()

A = 2.
w0 = 1. # rad/sec
A_1 = 5.
w0_1 = 3.
nin = 150
nout = 100000

x = rng.uniform(0, 10*np.pi, nin)

y = A * np.sin(w0*x) + A_1 * np.sin(w0_1*x)

w = np.linspace(0.01, 10, nout)

pgram = signal.lombscargle(x, y, w, normalize=True)

fig, (ax_t, ax_w) = plt.subplots(2, 1, constrained_layout=True)
ax_t.plot(x, y, 'b*')
ax_t.set_xlabel('Time [s])
```

```
ax_w.plot(w, pgram)
```

```
ax_w.set_xlabel('Angular frequency [rad/s]')
```

```
ax_w.set_ylabel('Normalized amplitude')
```

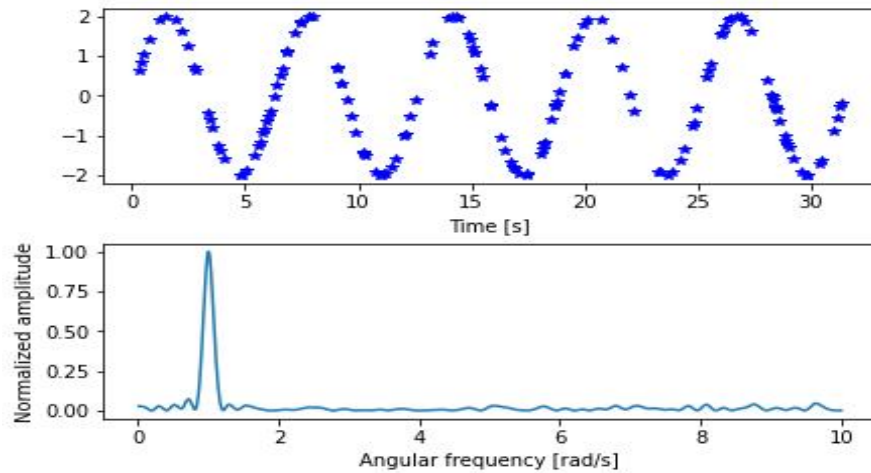


Fig 2.2.1 : Periodogram plot using lomb-scargle for a single signal.

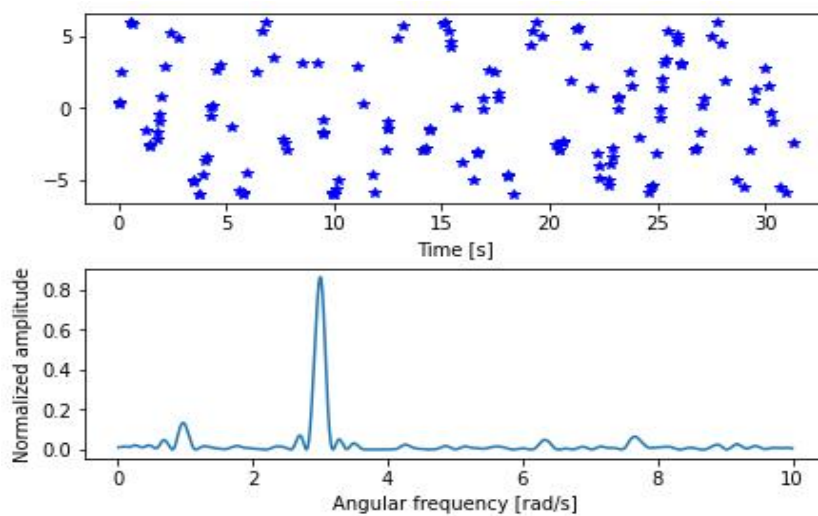


Fig 2.2.2 : Periodogram plot using lomb-scargle for the multiple signal.

Summary

The Welch periodogram and the Scargle periodogram are two commonly used techniques for analyzing the spectral content of time series data. The Welch periodogram is a modification of the standard periodogram that uses overlapping segments of the data to reduce the effects of noise and improve the estimation of the spectral density. Whereas the scargle periodogram is a method for analyzing unevenly sampled time series data, where the time intervals between observations may not be constant.

The welch works by dividing the time series into smaller segments, applying a window function to each segment to reduce spectral leakage, and then averaging the periodograms of the segments and The scargle works by fitting a sinusoidal model to the data at a range of frequencies and computing a statistic that measures the goodness of fit at each frequency. The resulting periodogram shows the strength of periodic signals in the data as a function of frequency. The choice between them depends on the nature of the data and the specific analysis goals.

References

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